SILENTHEAT: QUANTIFYING THE HUMID-HEAT BLIND SPOT IN INDIA'S HEATWAVE WARNINGS

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Abstract

India declares a heatwave only when the **dry-bulb** temperature crosses a fixed threshold: 40 °C on the plains or 37 °C on the coast, together with a 4.5 °C departure from normal. That declaration is what activates city and state Heat Action Plans. But the body's ability to shed heat is governed by **humid heat**, measured by the wet-bulb temperature (Tw). We built a fully reproducible R pipeline over 164,360 city-days of ERA5 reanalysis (10 Indian cities, 1981–2025). It applies the official IMD rule day by day and audits it against a physiological humid-heat criterion (daily maximum Tw ≥ 29 °C).

Of the **730 dangerous humid-heat days** we found, **703 (96.3%, Wilson 95% CI 94.7%–97.4%) were never IMD heatwave days**. Agreement between the two definitions is essentially zero (Cohen's κ -0.00–0.09). The miss rate is statistically the same across cities and between coastal and plains rules. It stays at 96–99% for every threshold we tried, including each city's own 95th/99th percentile. Meanwhile, danger days are increasing by **54% per decade** (negative-binomial IRR 1.54, 95% CI 1.34–1.77). Under a non-stationary GEV model, Chennai's 1-in-50-year humid-heat extreme of 1985 is now a **1-in-2.6-year** event. A simple logistic early-warning model gives a calibrated next-day probability with AUC 0.92, against 0.61 for the IMD rule's state. All analysis, figures, this report and an interactive Shiny dashboard are produced in R.

1. The gap

In practice. IMD's heatwave criteria are dry-bulb only. A 2023 Centre for Policy Research review of 37 Heat Action Plans across 18 states found that only 10 set locally defined thresholds. It also found that plans focus on dry heat and largely ignore humid heat and warm nights. IMD does publish an *experimental* Heat Index (NOAA-style, colour-coded green/yellow/orange/red), but it is guidance only. It does not trigger the heatwave declaration, and it gives no probabilities or return periods.

In the physiology. Laboratory work (Vecellio et al., 2022) puts the critical wet-bulb limit for young, healthy adults in warm-humid air at about 30–31 °C, well below the theoretical 35 °C. Observational work (Raymond et al., 2020) shows that extreme humid heat has more than doubled in frequency since 1979.

The unanswered question. How much physiologically dangerous humid heat falls outside India's official heatwave rule? Is that blind spot growing, and can probability modelling close it?

Research question	Statistical tools (all in R)
RQ1 What share of dangerous humid-heat days does the IMD rule miss?	Wilson / exact / year-block bootstrap CIs, Beta–Binomial posterior, χ^2 / Fisher, McNemar, Cohen's κ, two-proportion test
RQ2 Are danger days becoming more frequent?	Mann–Kendall + Sen's slope (own code), Poisson vs negative-binomial GLM, Wilcoxon, Welch t, KS, permutation test, two-way ANOVA + Tukey, Friedman
RQ3 How extreme can humid heat get, and is the tail shifting?	GEV maximum likelihood (own optimiser code), Gumbel LRT, delta-method + parametric bootstrap return levels, non-stationary GEV
RQ4 Do dangerous days cluster into spells?	Two-state Markov chain, χ^2 / G^2 dependence tests, geometric spell law, Monte Carlo simulation
RQ5 Can we warn a day ahead, with honest probabilities?	Logistic regression, temporal hold-out, Brier skill, ROC/AUC (DeLong), reliability diagram

2. Data

Daily ERA5 reanalysis (0.25°) was downloaded from the Open-Meteo archive API by `R/01_fetch_data.R`. The window is 1 Jan 1981 to 31 Dec 2025 for ten cities: seven judged by IMD's plains rule (Delhi, Lucknow, Ahmedabad, Nagpur, Kolkata, Bhubaneswar, Vellore) and three by the coastal rule (Chennai, Mumbai, Visakhapatnam). Variables: daily max/min temperature, daily max/mean wet-bulb temperature, and daily min/mean relative humidity. The dataset has no missing values.

Physics check. We recomputed wet-bulb temperature independently from (Tmax, RHmin) with Stull's (2011) formula. It correlates with ERA5's daily maximum Tw at r = 0.96–0.99 in every city (mean offset 0.51 °C), so the humid-heat signal is physically consistent.

Climatological normal. IMD judges departures from a station's normal. We estimate each city's normal Tmax over the WMO 1991–2020 base period with a 3-harmonic regression, $T_{max} = \beta_0 + \sum_{k=1}^{3}[a_k \sin(2\pi k\, d/365.25) + b_k \cos(2\pi k\, d/365.25)]$, with R^2 = 0.56–0.85.

The IMD rule as coded. Plains: Tmax ≥ 40 °C and departure ≥ 4.5 °C, or Tmax ≥ 45 °C. Coastal: Tmax ≥ 37 °C and departure ≥ 4.5 °C. Either way, the condition must hold on ≥ 2 consecutive days. We use one grid cell rather than IMD's two stations per subdivision, which makes our version **more generous to IMD**. Every miss rate below is therefore a conservative lower bound.

3. Exploratory analysis: why a dry-bulb rule cannot see humid heat

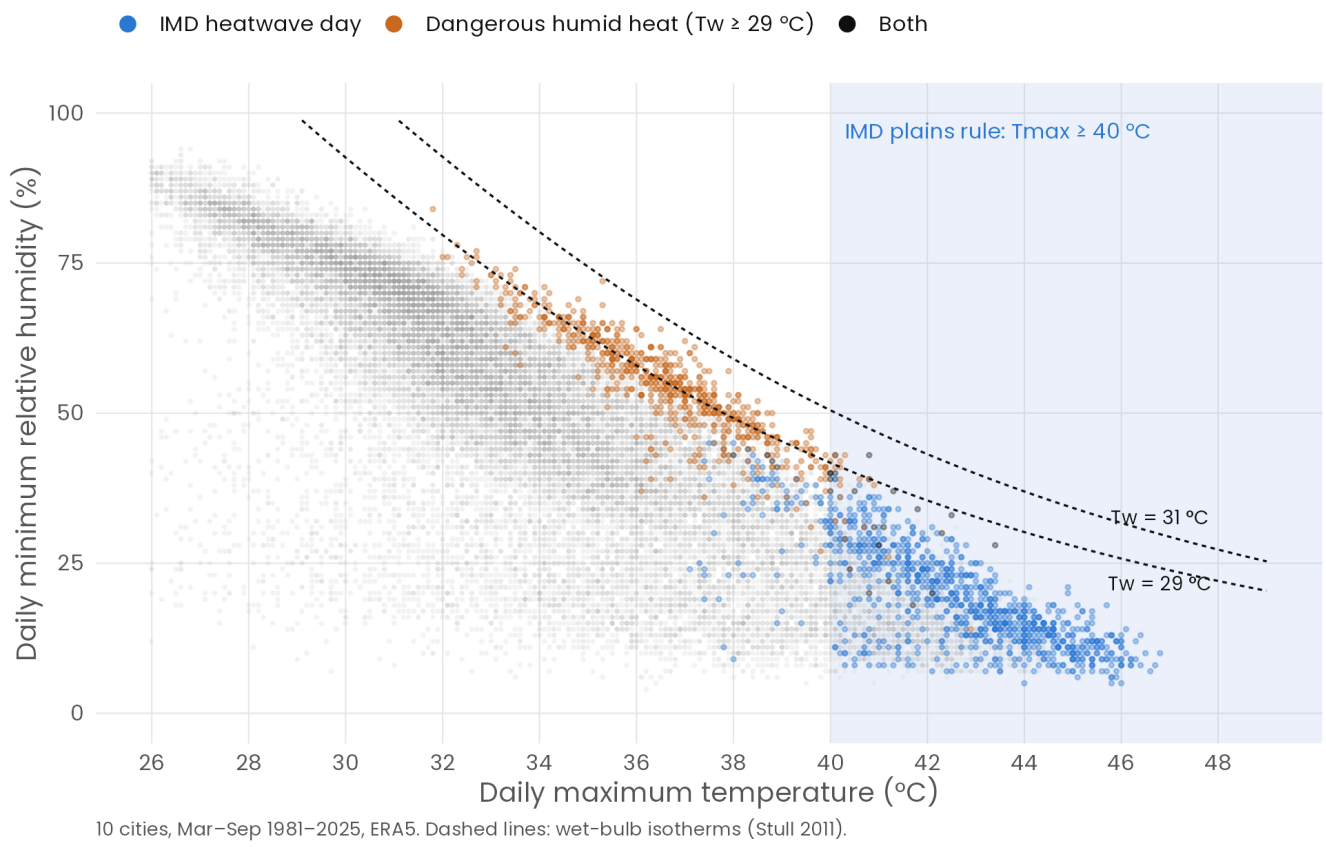


Figure 1. Each dot is a city-day (March–September). IMD heatwave days (blue) live in the hot-dry corner. Dangerous humid-heat days (orange) sit along the 29 °C wet-bulb isotherm at 32–39 °C and 40–70% humidity, a region the dry-bulb rule never enters. Only 27 of 990 IMD heatwave days were also humid-danger days.

Temperature and humidity are strongly **negatively** correlated on hot-season days in every city (Pearson r from -0.91 to -0.61, all $p < 0.001$; Spearman agrees). In other words, the hottest days are the driest days, so a threshold on temperature alone systematically selects dry heat. A multiple regression of daily max Tw on Tmax and RHmin (R^2 0.76–0.94) shows that +10 percentage points of humidity raises Tw by 1.5–2.1 °C at fixed temperature. In standardised terms, humidity is as important a driver as temperature.

City	IMD rule	Mean Tmax	Mean Tw max	SD Tw max	Max Tw	Skew	Danger d/yr	IMD HW d/yr	Silent d/yr
Delhi	Plains	34.66	23.63	3.78	29.9	-0.74	0.29	5.27	0.29
Lucknow	Plains	34.37	24.41	3.58	31.3	-0.91	0.69	6.07	0.67
Ahmedabad	Plains	35.21	24.15	2.97	28.7	-1.21	0.00	1.38	0.00
Nagpur	Plains	34.61	23.47	2.56	28.3	-0.88	0.00	3.98	0.00
Kolkata	Plains	32.73	26.61	1.97	31.0	-1.89	6.84	1.11	6.64
Bhubaneswar	Plains	33.49	26.54	1.49	30.9	-1.17	3.58	1.53	3.33
Vellore	Plains	35.12	24.30	1.32	29.5	-1.28	0.02	0.18	0.02
Chennai	Coastal	34.19	26.03	1.22	30.8	-0.39	1.42	0.13	1.40
Mumbai	Coastal	30.15	25.26	1.39	29.3	-1.38	0.04	0.40	0.04
Visakhapatnam	Coastal	32.34	26.39	1.30	30.3	-0.63	3.33	1.96	3.22

Table 1. Descriptive statistics, March–September 1981–2025 (°C; d/yr = days per year).

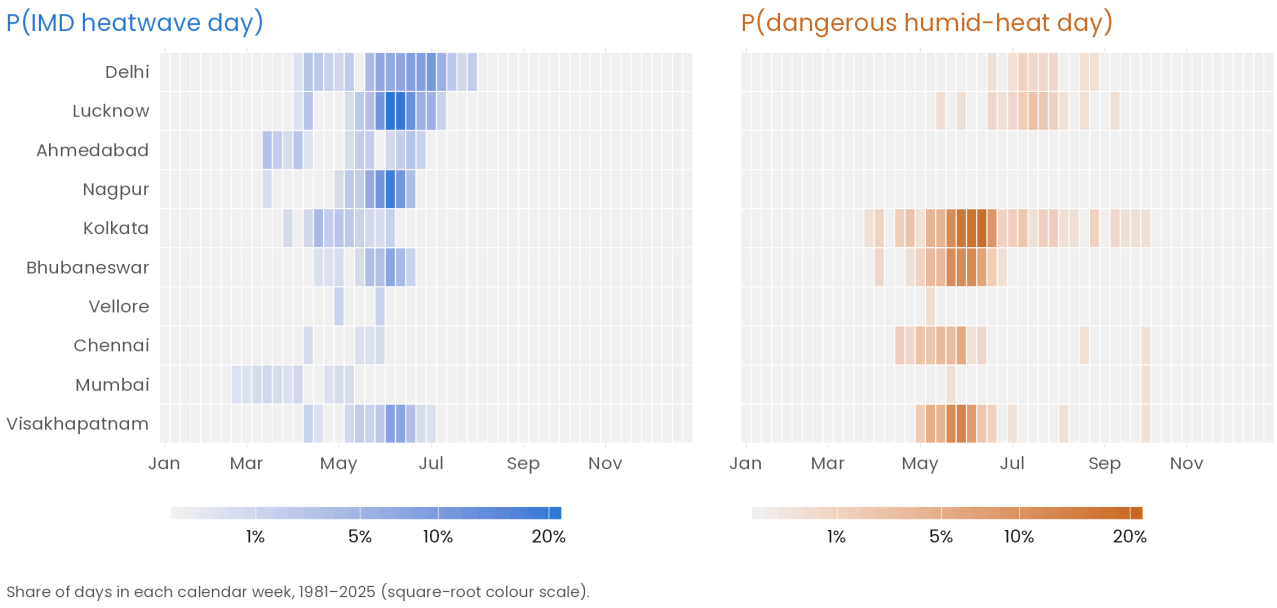


Figure 2. Probability of each event by calendar week. The IMD heatwave season (north and central India, April–June) and the humid-danger season (east coast and Kolkata, May–July) barely overlap.

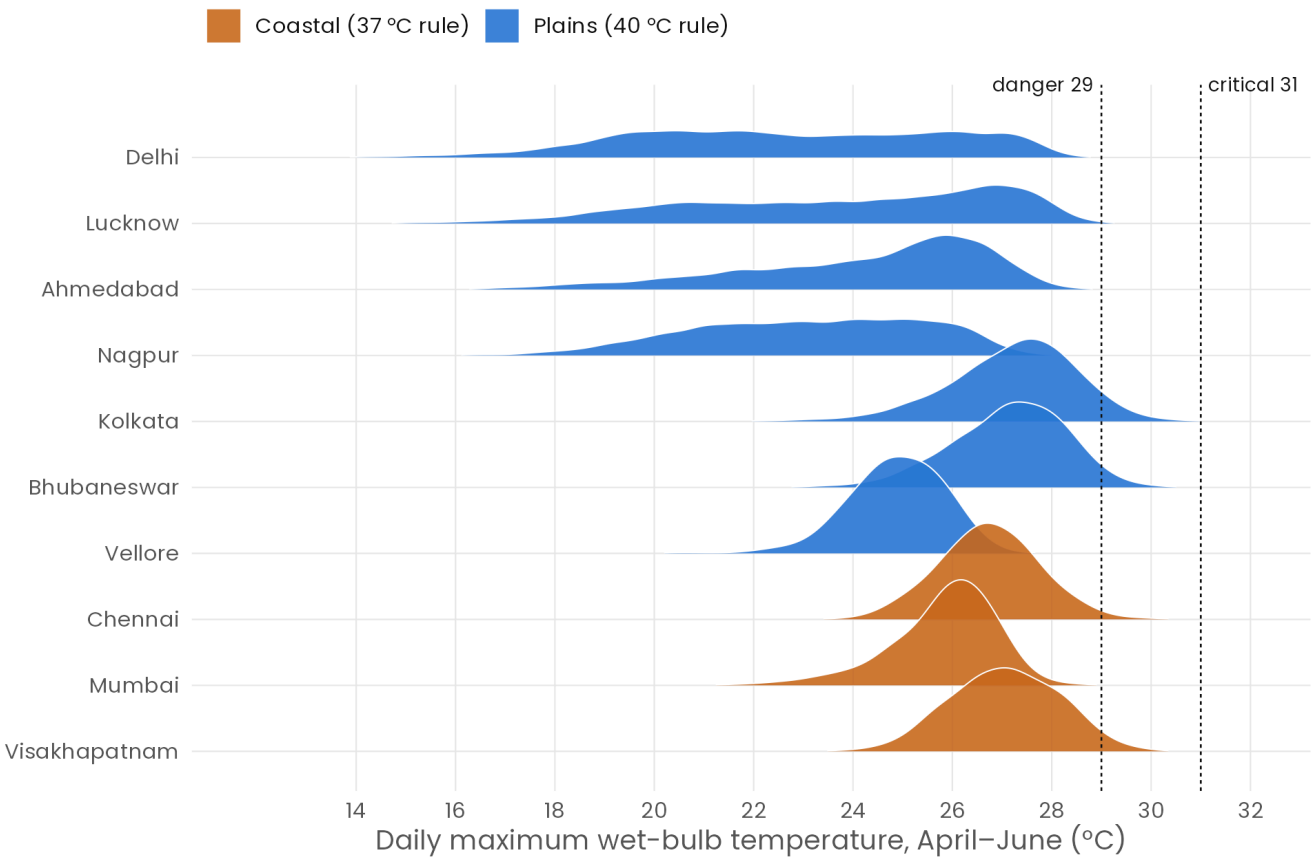


Figure 3. April–June distribution of daily maximum wet-bulb temperature. The distributions are left-skewed (skewness −0.4 to −1.9), with coastal and eastern cities pressed against the danger line.

4. Which probability model fits?

City	Distribution	log L	AIC	ΔAIC	KS D	Anderson–Darling
Chennai	Normal	-1746.335	3496.670	6.959	0.050	2.537
Chennai	Gamma	-1743.756	3491.512	1.801	0.047	2.178
Chennai	Weibull	-1904.186	3812.372	322.661	0.103	27.954
Chennai	Lognormal	-1742.856	3489.711	0.000	0.045	1.973
Delhi	Normal	-3048.830	6101.660	0.000	0.041	2.806
Delhi	Gamma	-3063.582	6131.164	29.504	0.045	3.746
Delhi	Weibull	-3053.921	6111.843	10.182	0.064	5.841
Delhi	Lognormal	-3074.157	6152.314	50.654	0.048	4.581
Kolkata	Normal	-1990.829	3985.658	0.000	0.052	2.991
Kolkata	Gamma	-2582.317	5168.633	1182.975	0.214	150.018
Kolkata	Weibull	-2025.139	4054.278	68.620	0.059	6.541
Kolkata	Lognormal	-2000.847	4005.694	20.036	0.060	4.480

Table 2. Maximum-likelihood fits to daily max Tw in May (`fitdistrplus`). No single family wins everywhere, and the Anderson–Darling statistics stay large. The bulk fits miss the tail that matters for health, which is why we model extremes directly with extreme-value theory (Section 7).

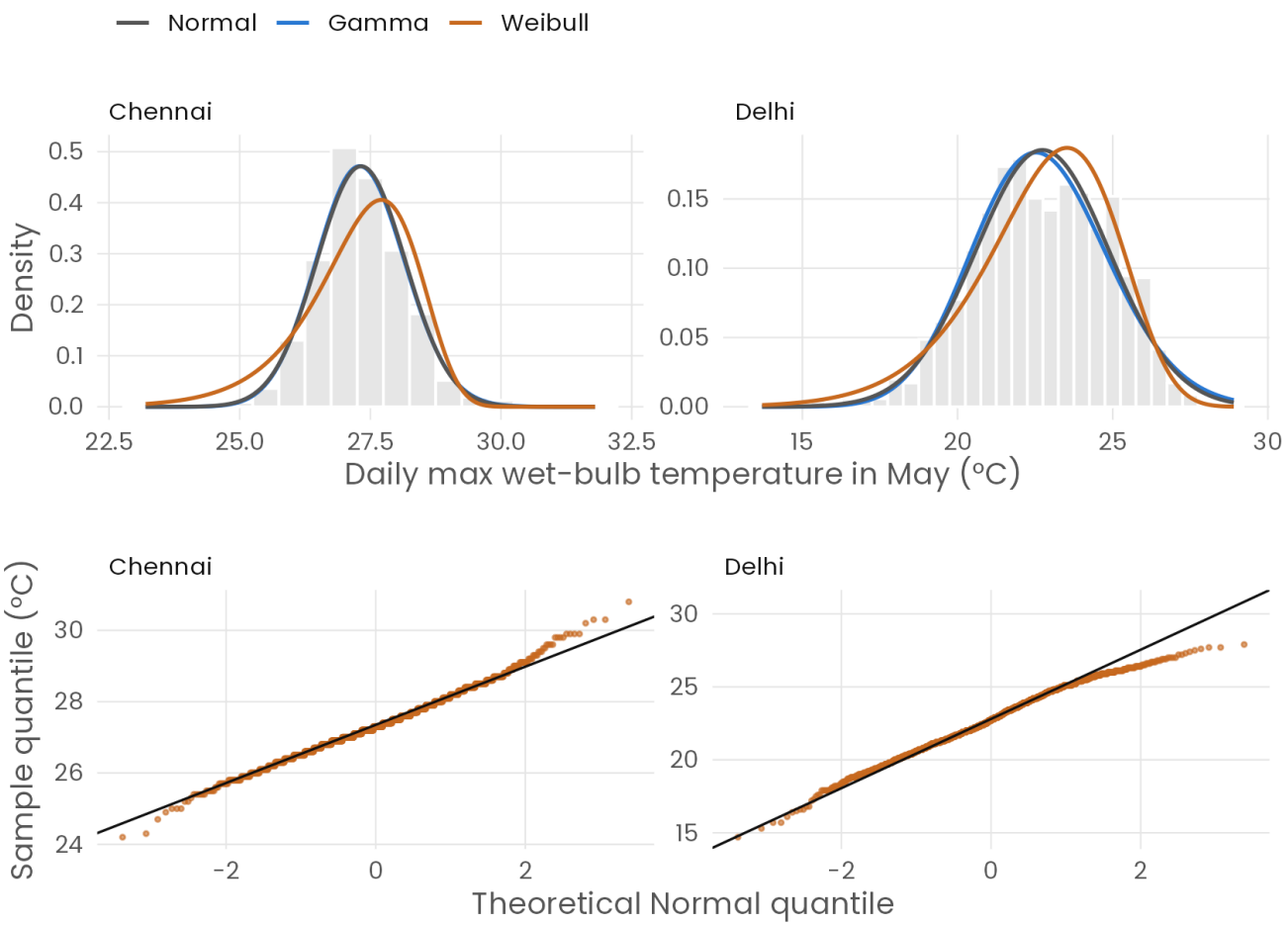


Figure 4. Histograms with fitted Normal, Gamma and Weibull densities, plus Normal Q–Q plots. Departures are concentrated in the tails.

For the **annual count of danger days**, the variance exceeds the mean by a factor of 2.0–9.0. The dispersion test rejects the Poisson model in every city ($p < 0.001$), and a negative binomial is preferred by AIC and by a boundary-corrected likelihood-ratio test. We therefore use the negative binomial for trend inference.

City	Mean	Variance	Var/mean	Dispersion p	AIC Poisson	AIC NegBin	LRT p
Bhubaneswar	3.58	21.52	6.02	< 0.001	338.65	217.27	< 0.001
Chennai	1.42	12.79	9.00	< 0.001	259.77	122.74	< 0.001
Delhi	0.29	0.57	1.99	< 0.001	70.80	59.37	< 0.001
Kolkata	6.84	51.09	7.46	< 0.001	411.51	271.87	< 0.001
Lucknow	0.69	1.76	2.56	< 0.001	121.17	104.07	< 0.001
Visakhapatnam	3.33	17.32	5.20	< 0.001	304.27	213.63	< 0.001

5. RQ1: The blind spot

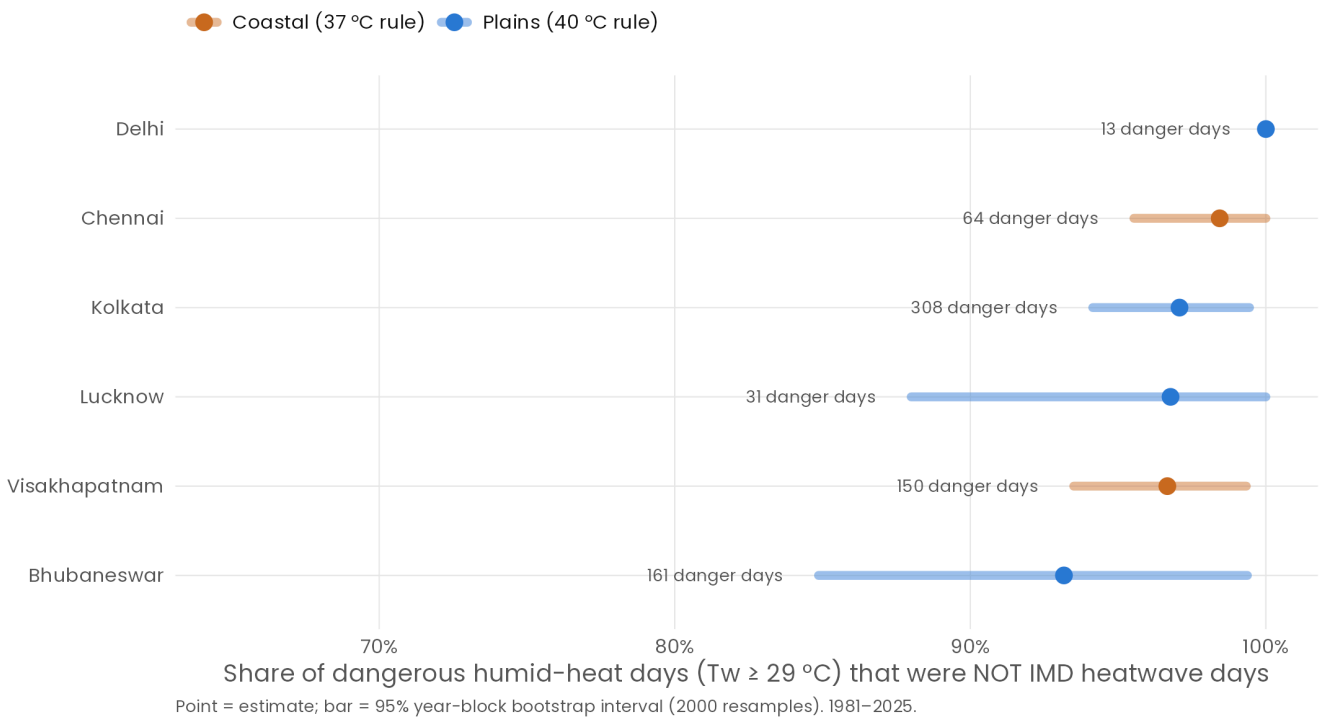


Figure 5. Share of danger days that were not IMD heatwave days, with 95% year-block bootstrap intervals. Whole years are resampled to respect day-to-day dependence.

City	Danger days	Missed	Miss rate	Wilson 95% CI	Bootstrap 95% CI	Bayes 95% CrI	Cohen κ	McNemar p	HI ≥ orange
Bhubaneswar	161	150	93.2%	88.2% – 96.1%	84.9% – 99.4%	88.2% – 96.1%	0.090	< 0.001	96%
Chennai	64	63	98.4%	91.7% – 99.7%	95.5% – 100.0%	91.7% – 99.6%	0.028	< 0.001	41%
Delhi	13	13	100.0%	77.2% – 100.0%	100.0% – 100.0%	76.8% – 99.8%	-0.002	< 0.001	46%
Kolkata	308	299	97.1%	94.5% – 98.5%	94.1% – 99.4%	94.5% – 98.4%	0.045	< 0.001	87%
Lucknow	31	30	96.8%	83.8% – 99.4%	88.0% – 100.0%	83.8% – 99.2%	0.003	< 0.001	45%
Visakhapatnam	150	145	96.7%	92.4% – 98.6%	93.5% – 99.3%	92.4% – 98.5%	0.036	< 0.001	81%

Table 3. Cities with at least 10 danger days. The Bayesian column uses a Beta(1,1) prior, giving a Beta(1 + missed, 1 + caught) posterior. "HI ≥ orange" is the share of danger days on which IMD's own experimental Heat Index reached orange or red.

- **Pooled:** 703 of 730 danger days were missed (96.3%).
- **Agreement:** Cohen's κ is close to 0 everywhere, so the two definitions measure *different hazards*. McNemar's test confirms that the disagreement is systematic ($p < 0.05$ in every city).
- **Homogeneity:** a χ^2 test of equal miss rates across cities (Monte Carlo $p = 0.33$) and a two-proportion test of coastal vs plains ($p = 0.52$) find no difference. The blind spot is a property of the rule, not of one region.
- **Robustness:** the pooled miss rate is 98%, 98%, 99%, 98%, 96%, 97% for thresholds Local P95, Local P99, Tw ≥ 27, Tw ≥ 28, Tw ≥ 29, Tw ≥ 30 respectively (Figure 6).
- **The fix is within reach:** IMD's own experimental Heat Index was orange or red on 81% of the missed days. A humidity-aware index already exists; it simply does not drive the declaration.

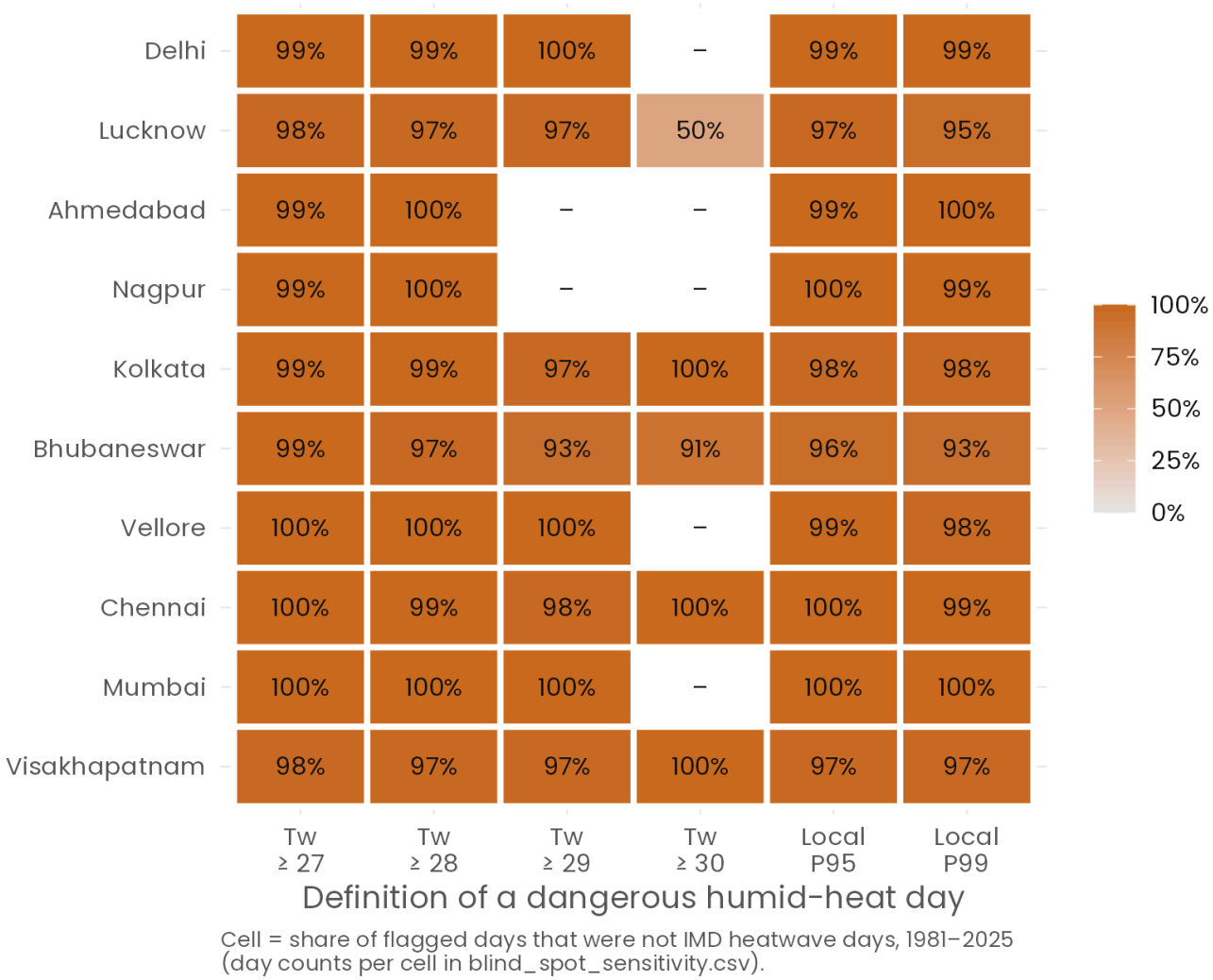


Figure 6. Miss rate under absolute and city-specific (local percentile) definitions of a dangerous day.

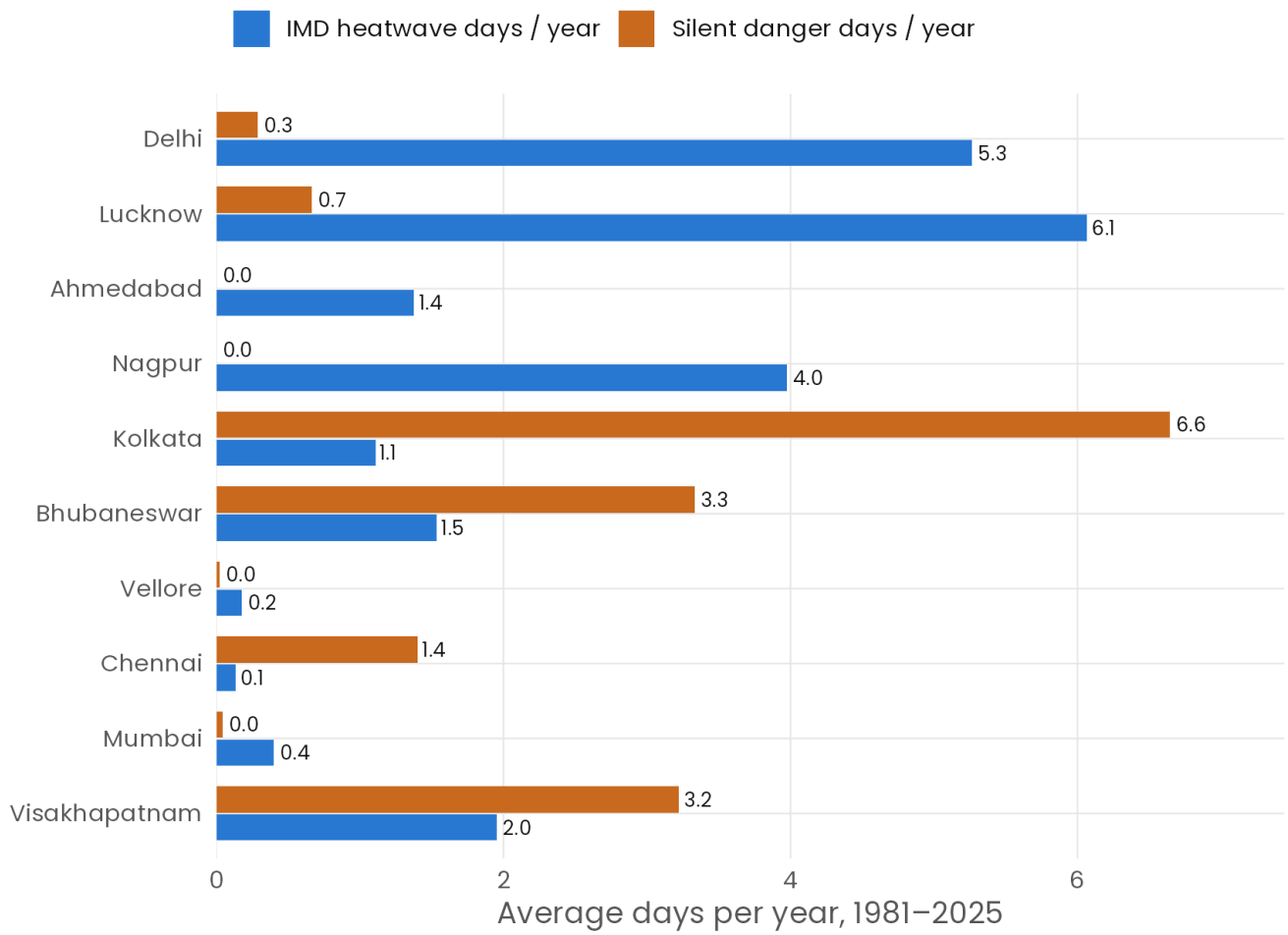


Figure 7. Average days per year. In Kolkata, Bhubaneswar, Visakhapatnam and Chennai, silent danger days outnumber official heatwave days.

6. RQ2: Is it getting worse?

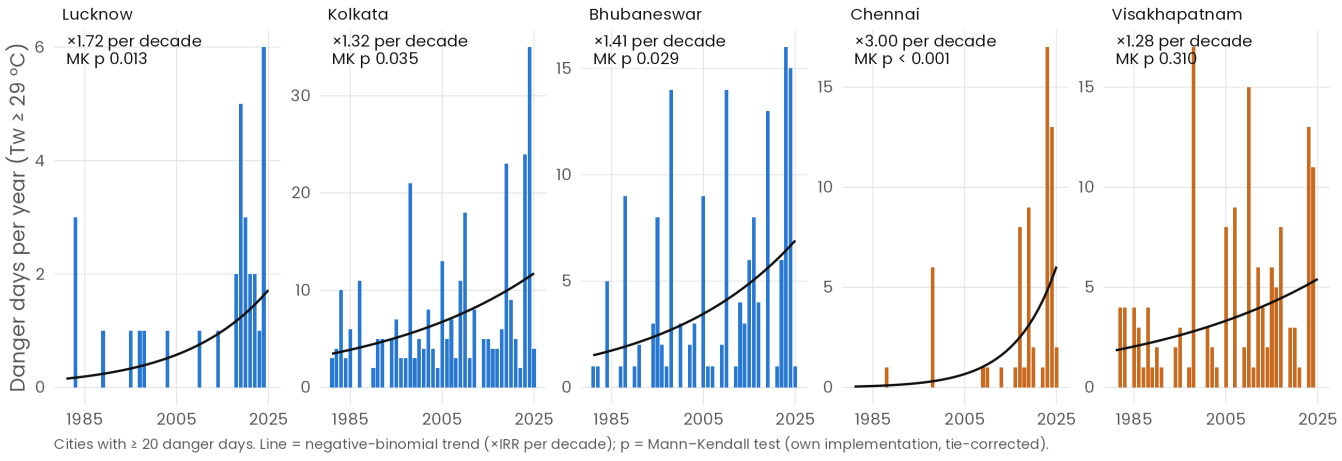
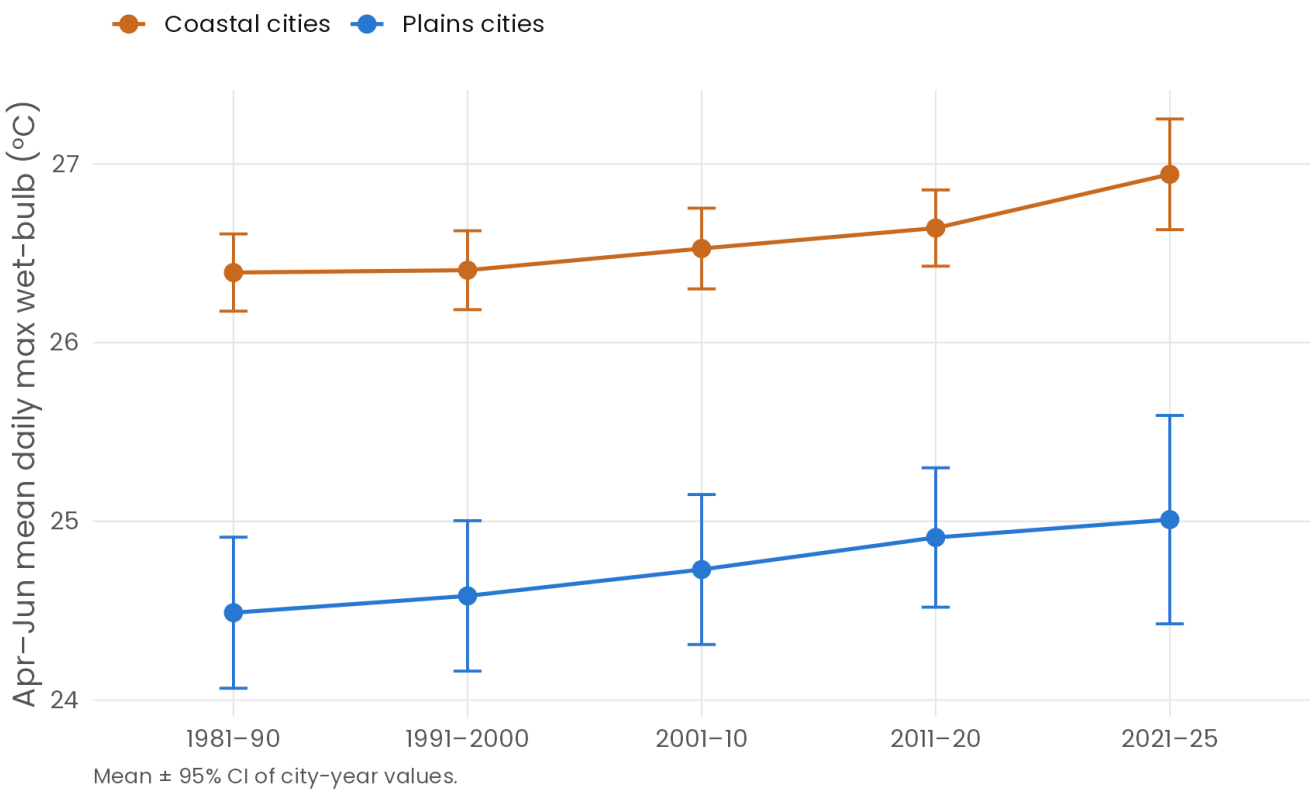


Figure 8. Annual danger days with per-city negative-binomial trend lines and Mann–Kendall p-values. The test is our own tie-corrected implementation, cross-checked against base R's `cor.test(method =`

`"kendall")`. Because most years have zero danger days, the Theil–Sen median slope is often exactly 0 even when the trend is significant. Sen slopes are therefore reported in `trend_mann_kendall.csv`, and the count model is used for effect sizes.

- **Count regression:** in a negative-binomial GLM with city fixed effects, danger days rise by a factor of **1.54 per decade** (95% CI 1.34–1.77, p = < 0.001). The Poisson model is badly overdispersed (Pearson dispersion 4.5; LRT p < 0.001).
- **Coastal vs plains:** fitting separate slopes, the increase is 1.76× per decade on the coast and 1.43× on the plains The coastal slope is steeper, but the difference is not significant (LRT for a common slope, p = 0.14).
- **Two-way ANOVA** of April–June mean Tw on city and decade: decade F = 20.0, p = < 0.001. Tukey HSD puts 2021–25 at **+0.53 °C** above 1981–90 (95% CI 0.33–0.73). Residuals are mildly non-normal (Shapiro–Wilk p = < 0.001) and heteroscedastic (Bartlett p = 0.001), so we confirmed the decade effect with the distribution-free **Friedman test**, blocking on city: p = < 0.001. A pooled Kruskal–Wallis test that ignores the city blocks is not significant (p = 0.19). That is a useful lesson in itself: between-city differences dwarf the decade signal unless the design accounts for them.



City	Danger d/yr 1981–2000	Danger d/yr 2006–2025	Permutation p	AMJ Tw 1981–2000	AMJ Tw 2006–2025	Welch p	KS p
Ahmedabad	0.00	0.00	1.000	23.963	24.464	< 0.001	< 0.001
Bhubaneswar	2.55	4.80	0.150	27.041	27.245	0.026	< 0.001
Chennai	0.35	2.85	0.033	26.555	26.950	0.003	< 0.001
Delhi	0.10	0.55	0.155	22.505	22.868	0.044	0.021
Kolkata	4.80	9.05	0.070	27.093	27.336	0.023	< 0.001
Lucknow	0.35	1.15	0.085	23.568	23.980	0.022	< 0.001
Mumbai	0.00	0.10	0.491	25.660	26.020	0.001	< 0.001
Nagpur	0.00	0.00	1.000	22.833	23.412	0.003	< 0.001
Vellore	0.00	0.05	1.000	24.749	24.998	0.019	< 0.001
Visakhapatnam	2.35	4.45	0.137	26.979	27.150	0.049	< 0.001

Table 4. Early vs recent era. The permutation test uses 10,000 relabellings of years. The KS test compares the full daily April–June distributions.

7. RQ3: Extremes and return levels (GEV)

For each city we fit the Generalised Extreme Value distribution to the 45 annual maxima of daily max Tw by maximising

$$\ell(\mu,\sigma,\xi) = -n\log\sigma - (1+1/\xi)\sum_i \log y_i - \sum_i y_i^{-1/\xi}, \quad y_i = 1+\xi\frac{z_i-\mu}{\sigma}$$

with our own `optim`-based code (`fit_gev()` in `R/00_setup.R`). Our estimates match `evd::fgev` to every printed digit. Return levels $z_T = \mu - \frac{\sigma}{\xi}\left[1-\{-\log(1-1/T)\}^{-\xi}\right]$ get delta-method and parametric-bootstrap (500 refits) intervals. A non-stationary model with $\mu(t) = \mu_0 + \mu_1 t$ (t in decades) is tested against the stationary one by a likelihood-ratio test.

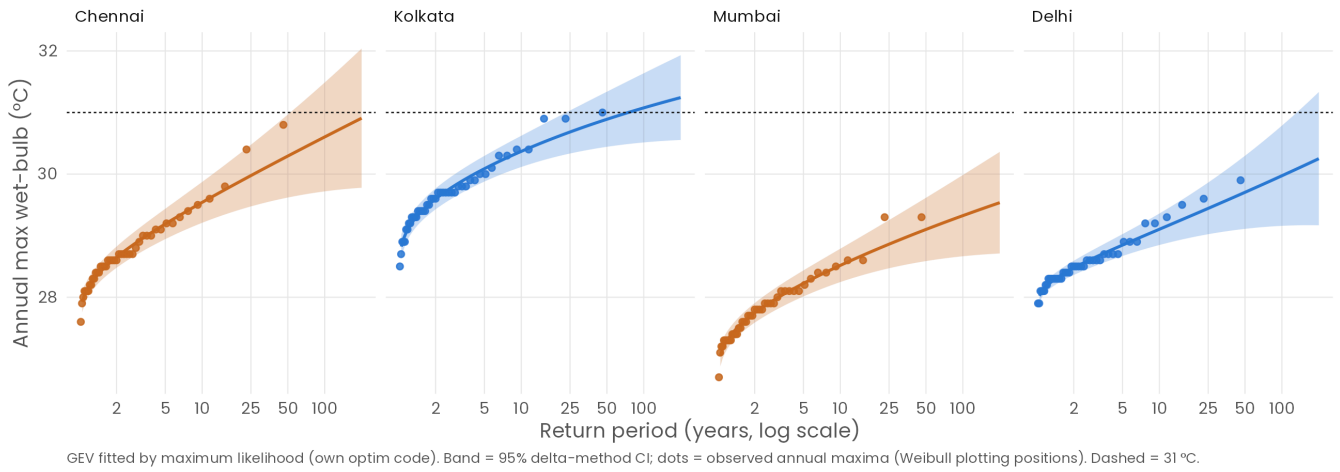


Figure 9. Return-level curves with 95% delta-method bands and observed annual maxima.

City	μ	σ	ξ (SE)	50-yr level	Bootstrap 95% CI	Trend °C/decade	Trend LRT p	1985's 50-yr event is now 1-in
Ahmedabad	27.88	0.42	-0.46 (0.09)	28.63	28.47 – 28.71	0.07	0.048	6.1
Bhubaneswar	29.00	0.29	0.38 (0.19)	31.61	30.25 – 34.99	0.04	0.237	41.8
Chennai	28.47	0.49	-0.02 (0.09)	30.29	29.59 – 31.16	0.25	< 0.001	2.6
Delhi	28.36	0.31	0.05 (0.11)	29.70	29.17 – 30.61	0.09	0.020	24.0
Kolkata	29.42	0.49	-0.15 (0.09)	30.89	30.40 – 31.39	0.15	0.007	6.8
Lucknow	28.59	0.31	0.22 (0.12)	30.51	29.63 – 32.36	0.06	0.133	36.8
Mumbai	27.58	0.46	-0.08 (0.09)	29.10	28.54 – 29.66	0.16	0.010	5.8
Nagpur	26.98	0.37	0.04 (0.14)	28.54	27.97 – 29.32	0.19	< 0.001	4.0
Vellore	26.51	0.46	0.04 (0.07)	28.44	27.66 – 29.51	0.18	< 0.001	12.3
Visakhapatnam	28.99	0.39	-0.13 (0.11)	30.20	29.77 – 30.60	0.05	0.320	19.3

Table 5. GEV parameters, 50-year return levels and the non-stationary shift.

The location of the humid-heat extreme distribution is rising significantly (LRT $p < 0.05$) in 7 of 10 cities. **Chennai's** shift is 0.25 °C per decade, so the annual maximum that was a 1-in-50-year event in 1985 is now expected about **once every 2.6 years**. For **Kolkata** the figure is once every 6.8 years. Under the trend model, the chance that Kolkata sees at least one ≥ 31 °C wet-bulb day in 2026–2035 is 35%.

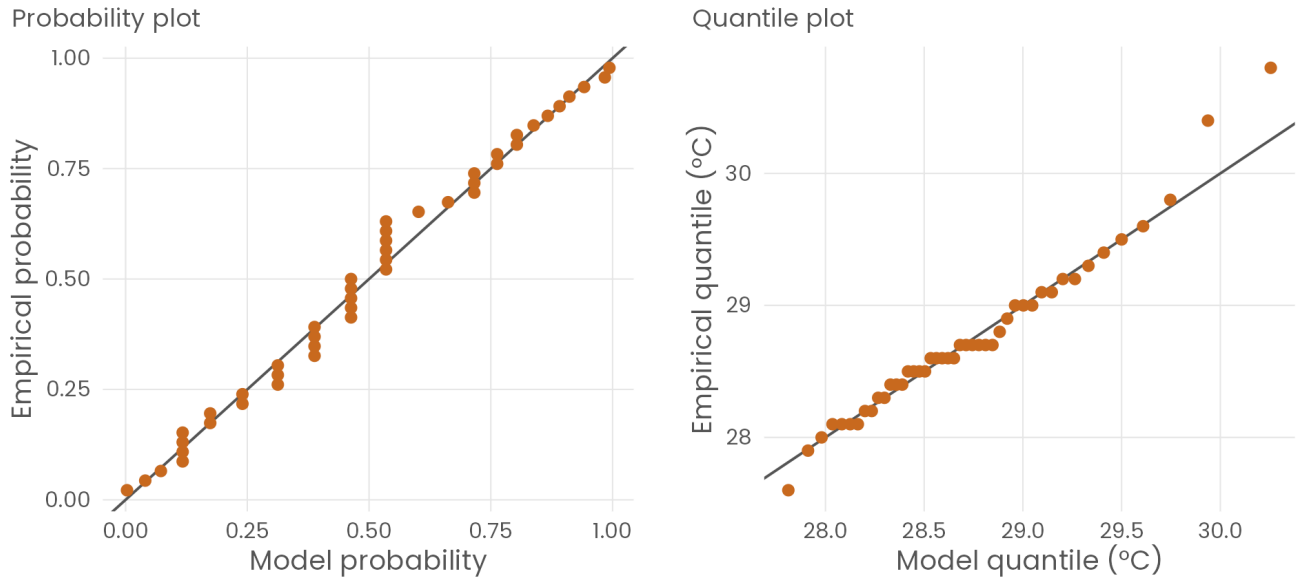
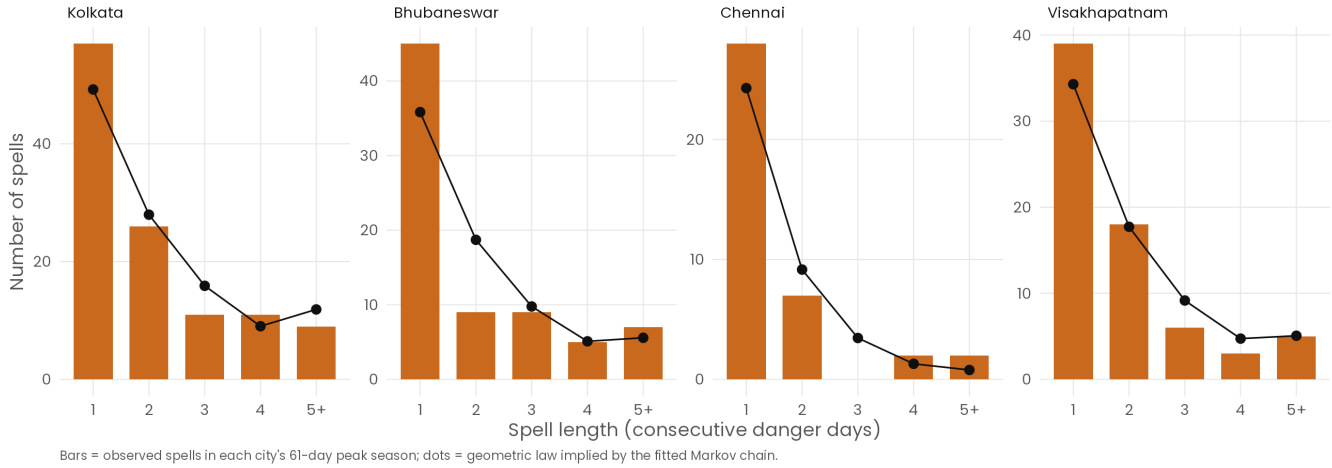


Figure 10. GEV probability and quantile plots for Chennai. Points lie close to the diagonal.

8. RQ4: Spells — a Markov chain view

Within each city's 61-day peak humid season, we model the day-to-day state (calm/danger) as a first-order, two-state Markov chain. For example, in Kolkata P(danger | danger yesterday) = 0.57, while P(danger | calm yesterday) = 0.05. Independence of consecutive days is decisively rejected in every city (χ^2 and G^2 $p < 0.001$; persistence odds ratios 27–41).

The chain implies geometrically distributed spell lengths with mean $1/(1 - p_{11})$. That matches the observed means (2.09, 1.59, 2.32, 2.07 vs 2.09, 1.61, 2.32, 2.07 days), and a χ^2 goodness-of-fit test does not reject the geometric law ($p = 0.10, 0.15, 0.41, 0.67$). The chain's stationary probability also reproduces the observed danger frequency almost exactly.



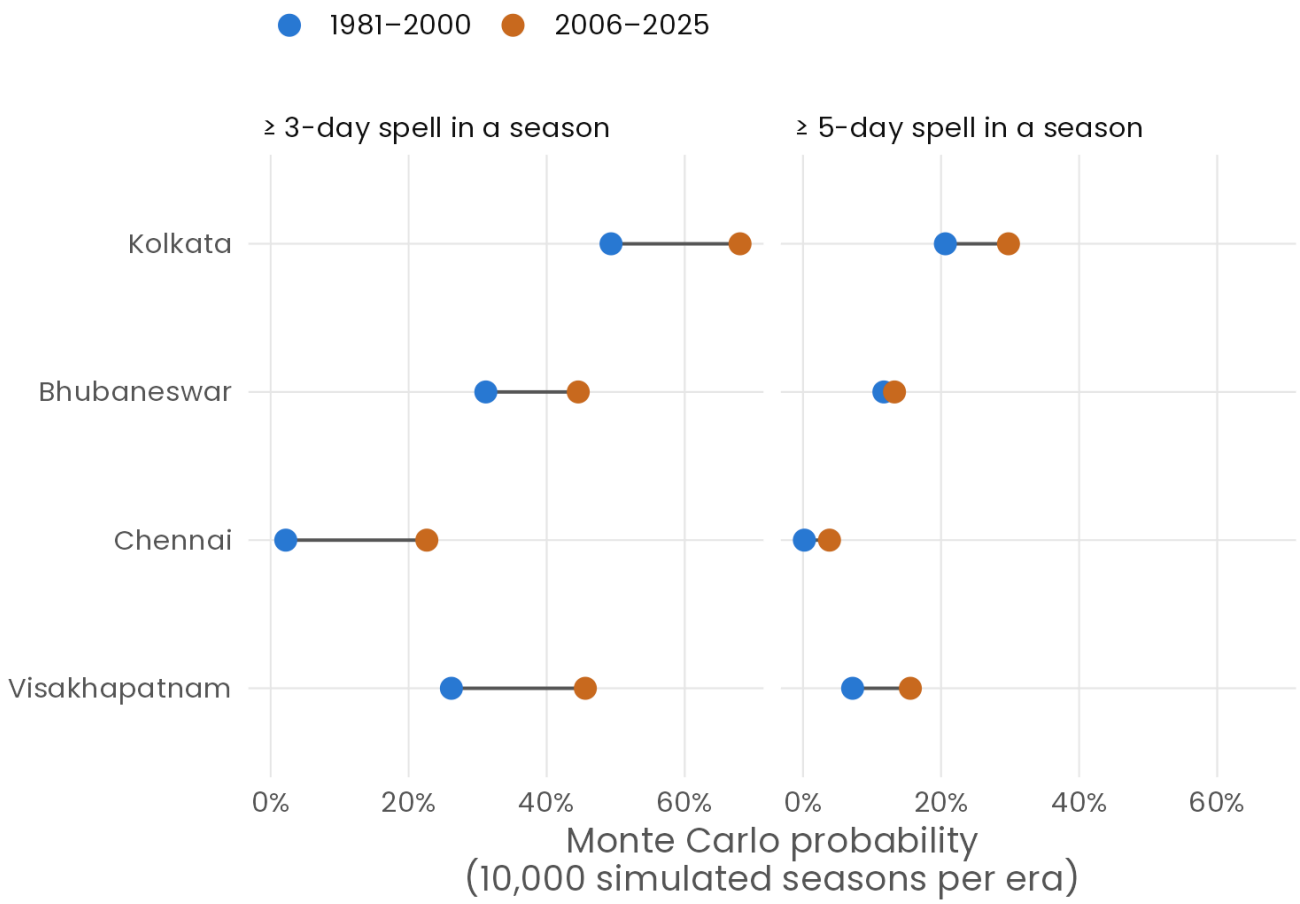


Figure 11. Monte Carlo (10,000 simulated seasons per era) using chains estimated separately for 1981–2000 and 2006–2025. In Chennai, the chance of a ≥ 3-day danger spell in a season rose from 2% to 23%. In Kolkata it rose from 49% to 68%.

9. RQ5: A probabilistic early warning

We fit a logistic regression for P(danger tomorrow) on 1981–2014 and test it on the unseen years 2015–2025. Predictors are today's max wet-bulb, its day-on-day change, Tmax, RHmin, Tmin, a decadal trend, and a city-specific seasonal cycle. It is compared with three baselines: climatology (seasonal base rate), persistence (P(tomorrow | today)), and the IMD-rule state (P(tomorrow | IMD heatwave today)).

Model	Brier	Brier skill vs climatology	Log-loss	AUC	AUC lo	AUC hi
Climatology	0.022	0.000	0.121	0.870	0.853	0.888
IMD-rule state	0.023	-0.059	0.121	0.611	0.577	0.644
Persistence	0.019	0.106	0.096	0.753	0.720	0.785
Logistic model	0.018	0.175	0.093	0.917	0.900	0.934

Table 6. Test-period skill. The AUC confidence intervals use the DeLong method. The logistic model beats persistence (DeLong Z = 10.4, p < 0.001).

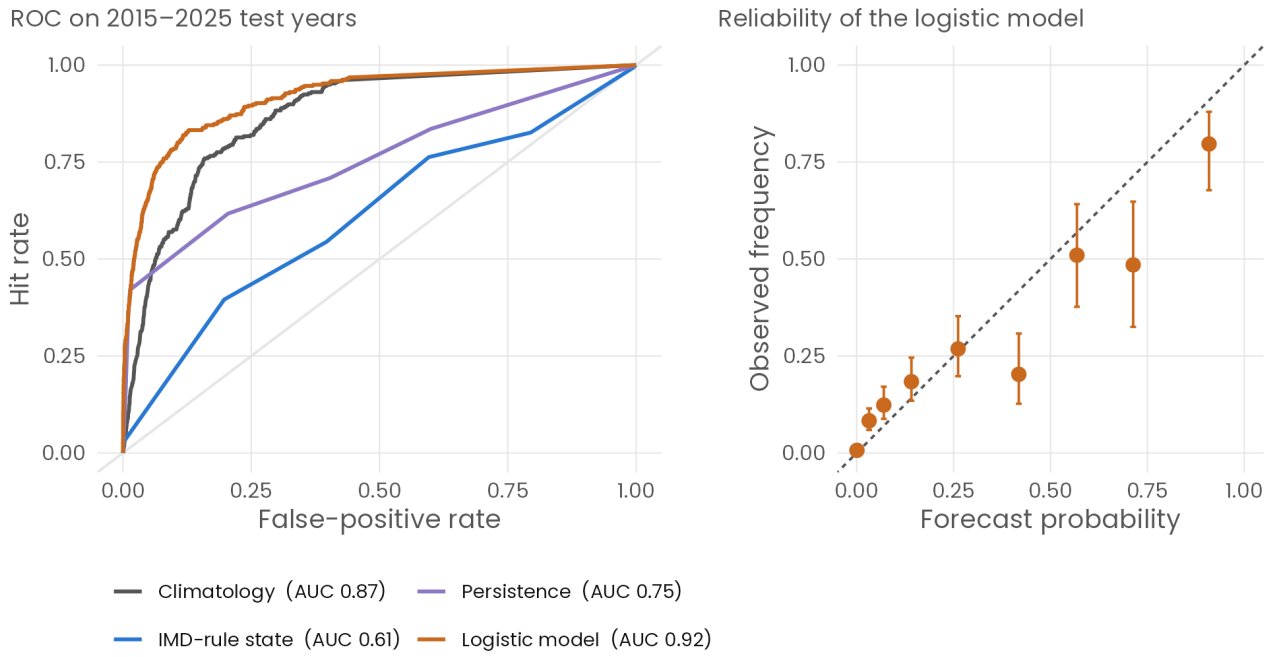


Figure 12. ROC curves and the reliability diagram (Wilson 95% intervals). Points near the diagonal mean the probabilities can be taken at face value.

Warning rule	Hits	False alarms	Misses	POD	FAR	CSI
IMD heatwave today	12	104	304	0.038	0.897	0.029
Danger today (persistence)	133	183	183	0.421	0.579	0.267
Logistic p ≥ 0.17 (max F1)	146	229	170	0.462	0.611	0.268
Logistic p ≥ 0.50	89	54	227	0.282	0.378	0.241

Table 7. Used as a warning, "IMD heatwave today" detects only 4% of next-day danger days. Each +1 °C of today's wet-bulb multiplies the odds of danger tomorrow by 11.3 (95% CI 7.5–17.1).

10. Discussion and recommendations

1. **Add a humid-heat trigger to heatwave declarations.** A wet-bulb or heat-index criterion would capture the ~96% of dangerous days the dry-bulb rule misses. IMD's experimental Heat Index already flagged 81% of them.
2. **Use local, probabilistic thresholds.** City-specific return levels (Table 5) and next-day probabilities (Section 9) turn a binary rule into a risk statement that a Heat Action Plan can act on.
3. **Prioritise the east coast and Gangetic West Bengal.** Kolkata, Bhubaneswar, Visakhapatnam and Chennai carry the largest silent burden, and danger days there are rising fastest.

Limitations. ERA5's 0.25° grid smooths local extremes and under-represents peak wet-bulb values (only 2 grid-days reached 31 °C). Mumbai's grid cell is strongly sea-moderated. Station data would raise every count, but the dry/humid separation is structural, not an artefact. We approximate IMD's two-station rule with one grid cell, which flatters IMD. Physiological thresholds come from studies of young, healthy adults, so risk for older or outdoor-working people starts lower. Daily dependence is handled with year-block resampling and Markov models, but some p-values from day-level tests remain approximate. GEV trend extrapolation beyond 2025 assumes the recent trend continues.

11. Conclusion

India's official heatwave rule and physiologically dangerous humid heat describe almost disjoint sets of days. Using a full probability-and-statistics workflow in R (distribution fitting, hypothesis tests, confidence and credible intervals, count regression, extreme-value theory, Markov chains, Monte Carlo and logistic forecasting), we showed three things. The blind spot is large (~96%), uniform across cities, robust to definitions, and growing. And it is fixable with a probabilistic, humidity-aware early warning.

Reproducibility

```
Rscript run_all.R                    # download → analysis → figures → this
report
Rscript -e 'shiny::runApp("app")'    # interactive dashboard
```

R 4.6.1; packages: dplyr, tidyr, ggplot2, patchwork, ggridges, scales, jsonlite, fitdistrplus, MASS, evd, pROC, shiny, bslib, knitr, markdown. The core statistical routines (GEV likelihood and return levels, Mann–Kendall/Sen, Wilson interval, Cohen's κ, Markov chain estimation and simulation, block bootstrap, permutation test, Stull wet-bulb, NWS heat index) are implemented from first principles in `R/00_setup.R` and the module scripts.

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Appendix: core R implementations

```
mann_kendall <- function(x, conf = 0.95) {
  x <- x[!is.na(x)]
  n <- length(x)
  ij <- which(upper.tri(matrix(0, n, n)), arr.ind = TRUE)  # pairs i < j
  i <- ij[, 1]; j <- ij[, 2]
  S <- sum(sign(x[j] - x[i]))
  ties <- table(x); ties <- ties[ties > 1]
  varS <- (n * (n - 1) * (2 * n + 5) - sum(ties * (ties - 1) * (2 * ties + 5)))
/ 18
  Z <- if (S > 0) (S - 1) / sqrt(varS) else if (S < 0) (S + 1) / sqrt(varS)
else 0
  slopes <- sort((x[j] - x[i]) / (j - i))
  N <- length(slopes)
  C <- qnorm(1 - (1 - conf) / 2) * sqrt(varS)
  lo <- slopes[max(1, floor((N - C) / 2))]
  hi <- slopes[min(N, ceiling((N + C) / 2) + 1)]
  data.frame(n = n, S = S, var_S = varS, Z = Z, p_value = 2 * pnorm(-abs(Z)),
             tau = S / (n * (n - 1) / 2), sen_slope = median(slopes),
             sen_lo = lo, sen_hi = hi)
}
```

```
gev_nll <- function(par, z, t = NULL) {
  # par = (mu0, [mu1], log_sigma, xi); location linear in covariate t if given
  if (is.null(t)) { mu <- par[1]; sigma <- exp(par[2]); xi <- par[3] }
  else            { mu <- par[1] + par[2] * t; sigma <- exp(par[3]); xi <-
par[4] }
  y <- 1 + xi * (z - mu) / sigma
  if (any(y <= 0) || sigma <= 0) return(1e10)
  if (abs(xi) < 1e-6) {
    s <- (z - mu) / sigma
    return(length(z) * log(sigma) + sum(s) + sum(exp(-s)))
  }
  length(z) * log(sigma) + (1 + 1 / xi) * sum(log(y)) + sum(y^(-1 / xi))
}

fit_gev <- function(z, t = NULL) {
  start <- c(mean(z) - 0.45 * sd(z), log(0.78 * sd(z)), -0.1)
  if (!is.null(t)) start <- c(start[1], 0, start[2:3])
  best <- NULL
  for (xi0 in c(-0.3, -0.1, 0.1)) {
```

```r
    s <- start; s[length(s)] <- xi0
    f <- optim(s, gev_nll, z = z, t = t, method = "Nelder-Mead",
               control = list(maxit = 5000, reltol = 1e-12))
    f <- optim(f$par, gev_nll, z = z, t = t, method = "BFGS",
               control = list(maxit = 1000, reltol = 1e-12))
    if (is.null(best) || f$value < best$value) best <- f
  }
  p <- best$par
  # Report on the natural scale and get the observed information there
  nat <- p; nat[length(nat) - 1] <- exp(p[length(p) - 1])
  nll_nat <- function(q) { q2 <- q; q2[length(q2) - 1] <- log(q[length(q) - 1])
                           gev_nll(q2, z = z, t = t) }
  H <- optimHess(nat, nll_nat)
  V <- tryCatch(solve(H), error = function(e) MASS::ginv(H))
  nm <- if (is.null(t)) c("mu", "sigma", "xi") else c("mu0", "mu1", "sigma",
"xi")
  names(nat) <- nm; dimnames(V) <- list(nm, nm)
  list(par = nat, vcov = V, se = sqrt(pmax(diag(V), 0)), loglik = -best$value,
       n = length(z), conv = best$convergence)
}

# Return level for return period T (years) and its delta-method standard error
gev_return_level <- function(fit, T, mu_override = NULL) {
  mu <- if (is.null(mu_override)) fit$par["mu"] else mu_override
  sigma <- fit$par["sigma"]; xi <- fit$par["xi"]
  yp <- -log(1 - 1 / T)
  z  <- mu - sigma / xi * (1 - yp^(-xi))
  grad <- c(1,
            -(1 - yp^(-xi)) / xi,
            sigma * (1 - yp^(-xi)) / xi^2 - sigma / xi * yp^(-xi) * log(yp))
  se <- sqrt(as.numeric(t(grad) %*% fit$vcov[c("mu", "sigma", "xi"),
                                             c("mu", "sigma", "xi")] %*%
grad))
  data.frame(T = T, level = unname(z), se = se)
}
```